

Grant Yang

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(408) 568-4549

SKILLS

- **Languages:** Python, C, C++, Rust, Java, TypeScript
- **Technology:** Numpy, PyTorch, OpenGL, ReactJS, Git, Linux/Unix, Networking
- **Misc:** Wolfram Mathematica, Information Theory, Linear Algebra, Functional Programming

EDUCATION

- **University of Washington** Seattle, WA
Bachelor of Science in Electrical and Computer Engineering 2025 – 2029 (expected)
 - **Coursework:** MATH 208: Linear Algebra, CSE 123: Computer Programming III
- **Harker Upper School** San Jose, CA
4.6/4.8 GPA 2021 – 2025

EXPERIENCE

- **UW Math AI Lab - Algebraic Geometry (Winter 2026):** Used Lean to formalize results on monogenic extensions of regular local rings (with Prof. Bianca Viray and Bryan Boehnke). Produced two Mathlib pull requests: [#36421](#) and [#37527](#). [uw-math-ai/monogenic-exetensions](#)
- **UW Lean Hackathon - 1st Place:** With Leopold Mayer and Brian Nugent, wrote a tactic `cat_rw` (Category rewrite) to enable automatic rewriting using isomorphisms as if they were equalities, streamlining the workflows of mathematicians doing category theory with Lean. [leomayer1/cat-rw](#) for [uw2026leanhackathon.github.io](#).

PROJECTS

Underlines are clickable links.

- **C++ Chess Engine:** Designed a UCI chess engine based on alpha-beta search that achieved a 1600 Lichess rating. Utilized optimizations including “magic bitboards,” multithreading, and caching to search over 2M positions per second.
- **Custom LLM:** Implemented a transformer from scratch in PyTorch. Trained a 40M parameter model on my text messages, and performed a scaled-up training run on UW’s Hyak supercomputer of a 150M parameter model.
- **4-bit Breadboard Computer:** Built a functional 4-bit computer from 7400-series integrated circuit chips. Programmed the computer with a custom Instruction Set Architecture (ISA) to output the Fibonacci numbers.
- **Robot Control:** Used PID to drive a robot with Mecanum wheels, and implemented Jacobian inverse kinematics to plan motions for a robot arm with 5 degrees of freedom.
- **Pascal Compiler:** Designed and implemented a compiler for a custom Pascal dialect that targets MIPS32 assembly. Demonstrated complex programs on a MIPS32 emulator.